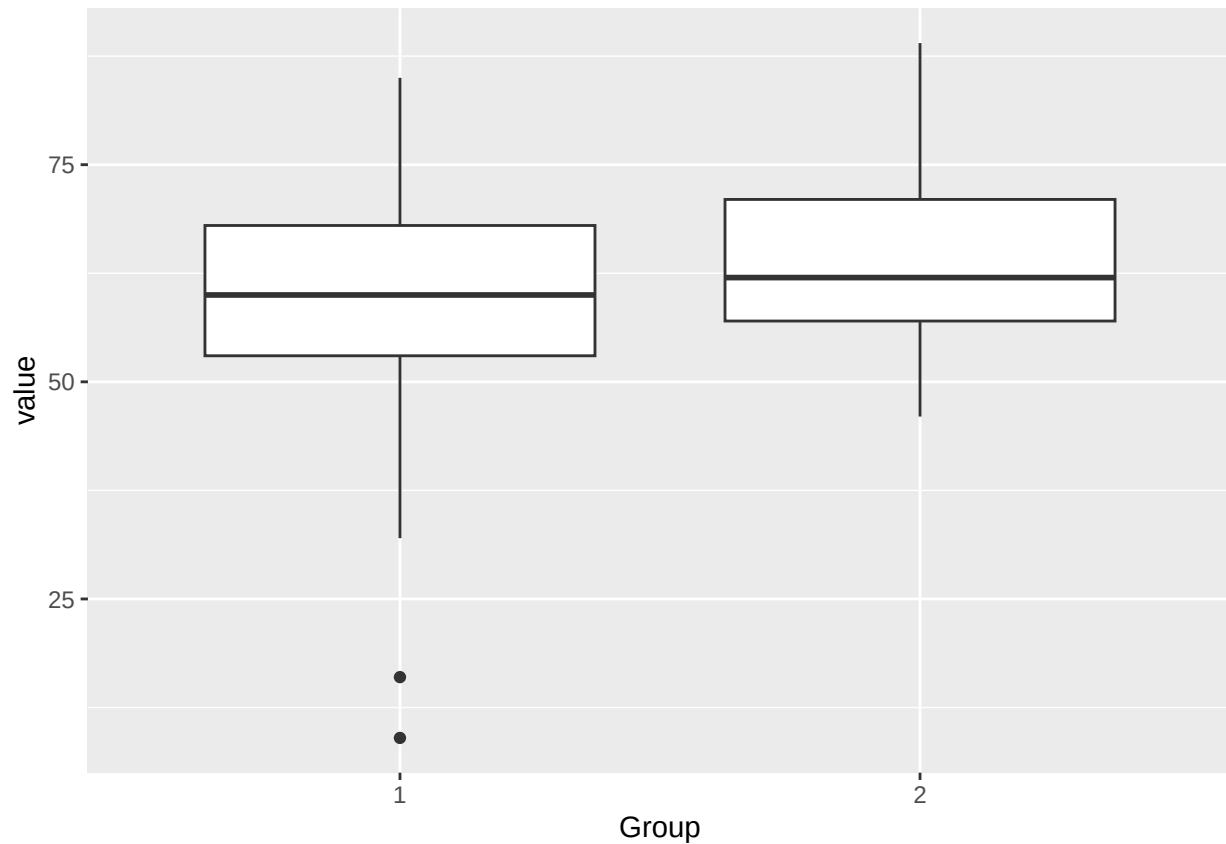


HW#3 FInal

2025-10-23

```
#setwd("cloud/project")
text1<- read.csv("TextMessages.csv", header = TRUE)

library(reshape)
text2<-melt(text1,id=c("Participant","Group"), measured=c("Baseline","Six_months"))
#
text2$Group<- as.factor(text2$Group)
#install.packages("ggplot2")
library(ggplot2)
ggplot(text2,aes(x=Group,y=value)) + geom_boxplot()
```



```
library(readr)
TextMessages_for_HW3 <- read_csv("TextMessages.csv")

nrow(TextMessages_for_HW3)
```

```
## [1] 50
```

```

#50 rows of data
names(TextMessages_for_HW3)

## [1] "Group"      "Baseline"    "Six_months"  "Participant"
#[1] "Group" "Baseline" "Six_Months" "Participant"

#install.packages("pastecs")

library(pastecs)

round( stat.desc(TextMessages_for_HW3$Six_months), 2)

##      nbr.val      nbr.null      nbr.na      min      max      range
##      50.00      0.00      0.00      9.00      79.00      70.00
##      sum      median      mean      SE.mean CI.mean.0.95      var
##      2870.00      60.50      57.40      1.97      3.96      194.12
##      std.dev      coef.var
##      13.93      0.24

#

round( stat.desc(TextMessages_for_HW3$Baseline), 2)

##      nbr.val      nbr.null      nbr.na      min      max      range
##      50.00      0.00      0.00      46.00      89.00      43.00
##      sum      median      mean      SE.mean CI.mean.0.95      var
##      3261.00      64.50      65.22      1.51      3.03      113.52
##      std.dev      coef.var
##      10.65      0.16

by(TextMessages_for_HW3$Six_months, TextMessages_for_HW3$Group, stat.desc)

## TextMessages_for_HW3$Group: 1
##      nbr.val      nbr.null      nbr.na      min      max      range
##      25.0000000      0.0000000      0.0000000      9.0000000      78.0000000      69.0000000
##      sum      median      mean      SE.mean CI.mean.0.95      var
##      1324.0000000      58.0000000      52.9600000      3.2662313      6.7411700      266.7066667
##      std.dev      coef.var
##      16.3311563      0.3083678
## -----
## TextMessages_for_HW3$Group: 2
##      nbr.val      nbr.null      nbr.na      min      max      range
##      25.0000000      0.0000000      0.0000000      46.0000000      79.0000000      33.0000000
##      sum      median      mean      SE.mean CI.mean.0.95      var
##      1546.0000000      62.0000000      61.8400000      1.8820910      3.8844450      88.5566667
##      std.dev      coef.var
##      9.4104552      0.1521742

#

by(TextMessages_for_HW3$Baseline, TextMessages_for_HW3$Group, stat.desc)

## TextMessages_for_HW3$Group: 1
##      nbr.val      nbr.null      nbr.na      min      max      range

```

```
##      25.000000      0.000000      0.000000      47.000000      85.000000      38.000000
##      sum      median      mean      SE.mean CI.mean.0.95      var
## 1621.000000    64.000000    64.840000      2.135946      4.408377    114.056667
##      std.dev      coef.var
##      10.679732      0.164709
```

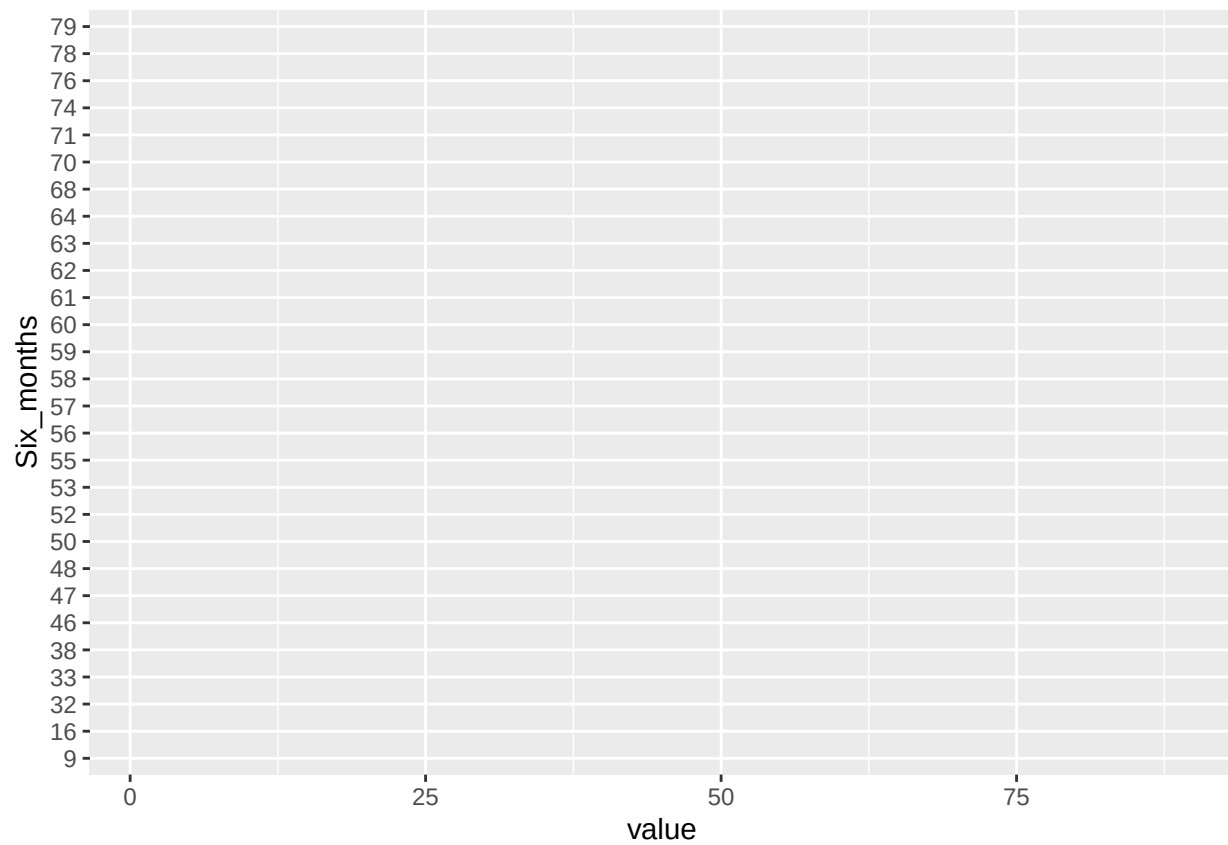
```
## -----
## TextMessages_for_HW3$Group: 2
##      nbr.val      nbr.null      nbr.na      min      max      range
##      25.0000000    0.0000000    0.0000000    46.0000000    89.0000000    43.0000000
##      sum      median      mean      SE.mean CI.mean.0.95      var
## 1640.000000    65.0000000    65.6000000      2.1671794      4.4728385    117.4166667
##      std.dev      coef.var
##      10.8358971      0.1651814
```

```
messages<- read.csv("TextMessages.csv", header = TRUE)
#install.packages("reshape")

library(reshape)
text3<-melt(messages,id=c("Participant","Six_months"), measured=c("Baseline","Group"))
#
text3$Six_months<- as.factor(text3$Six_months)

#install.packages("ggplot2")
library(ggplot2)
#Load the package and all of its dependencies

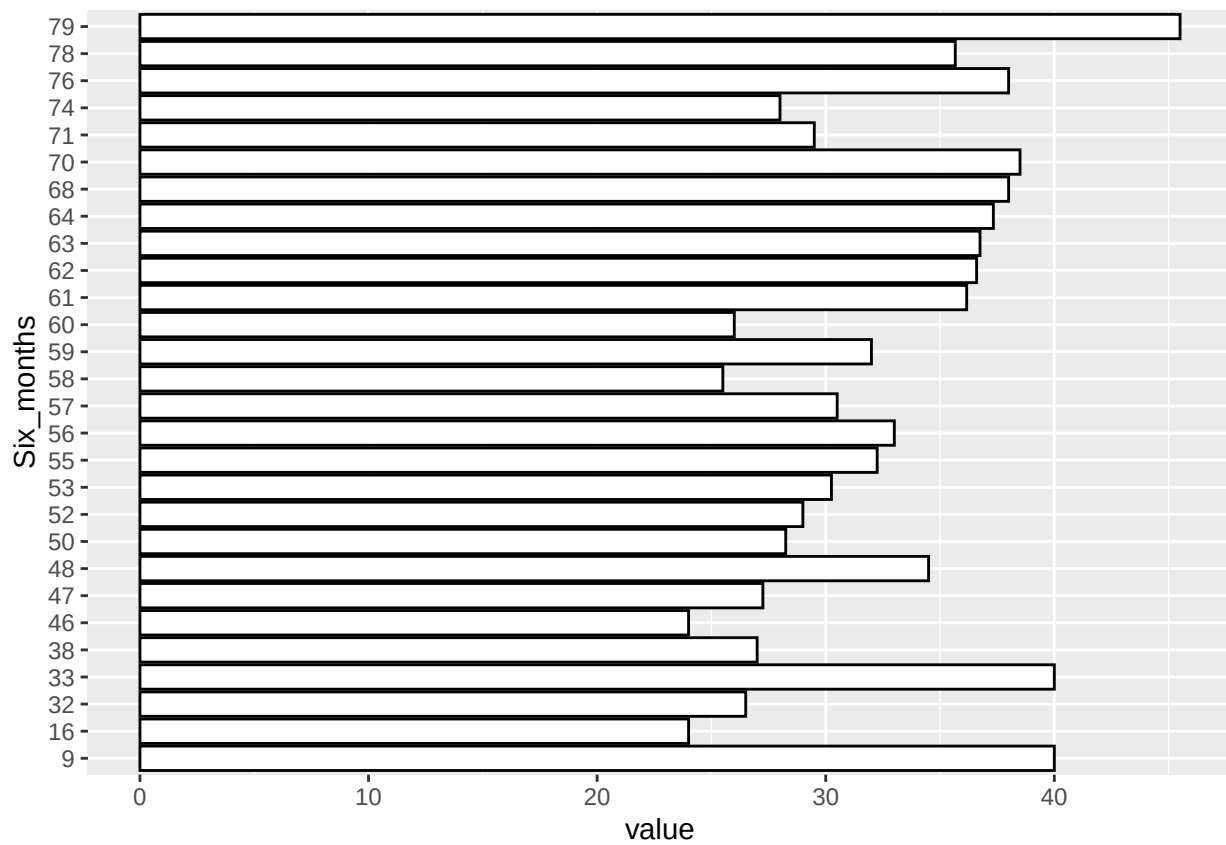
barWithErrors_valuebySix_months <- ggplot(text3, aes(value, Six_months))
barWithErrors_valuebySix_months
```



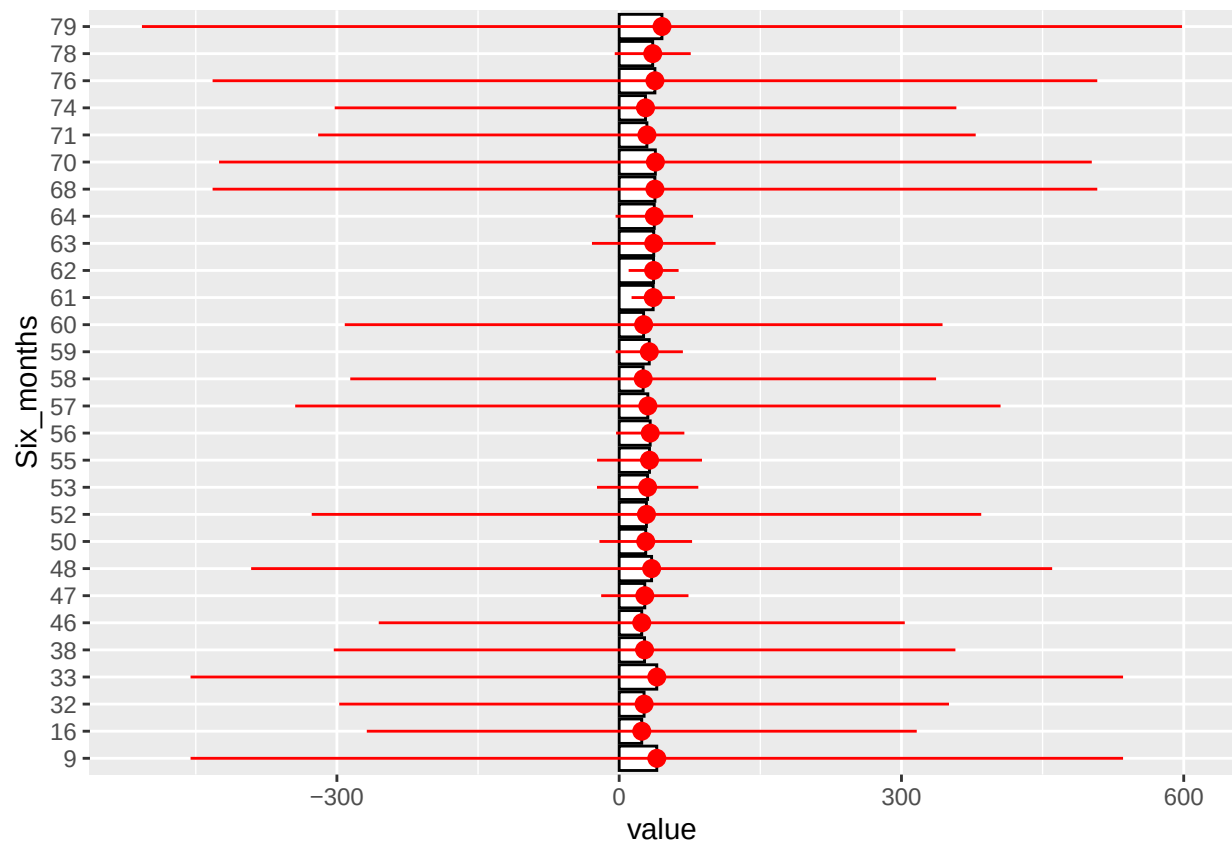
```
#install.packages("Hmisc")
```

```
library(Hmisc)
```

```
barWithErrors_valuebySix_months + stat_summary(fun. = mean, geom = "bar", fill =  
                                              "White", colour = "Black")
```



```
barWithErrors_valuebySix_months + stat_summary(fun= mean, geom = "bar", fill =
                                         "White", colour = "Black") + stat_summary(fun.data = m
                                         "pointrange")
```



```
barWithErrors_valuebySix_months + stat_summary(fun = mean, geom = "bar", fill =
                                         "Green", colour = "Blue") + stat_summary(fun.data = me
                                         "pointrange")
```

